

Machine learning, and in particular deep learning, is the backbone of most modern AI systems.

Traditional programming relies on explicitly defined rules and logic written by developers to solve problems, making it ideal for tasks with clear, predictable outcomes.

→ Supervised Learning

It's a machine learning technique that uses labeled data sets to train AI models to identify the underlying patterns and relationships. The goal is ^{of the} ~~for~~ process is to create a model that can predict correct outputs on a new real world data.

→ Unsupervised Learning

It uses machine learning algorithms to analyze and cluster unlabeled data sets. These algorithms discover hidden patterns, or data groupings without the need for human intervention.

→ Reinforcement Learning

It's a type of machine learning process in which autonomous agents learn to make decisions by interacting with their environment. In this learning agents learn to perform task by trial and error in the absence of any guidance from human user.

13. Deep Learning is a further subset of ML that uses deep neural networks with multiple layers of interconnected nodes to model complex patterns in large datasets.

